Markscheme — 9702/44/M/J/26 · Q5

Ball on a spring — velocity–displacement graph & resonance

(a)(i) [3 marks]

Draw a smooth closed loop (ellipse) enclosing the origin, cutting the x-axis at x = **±2.5 cm** (and nowhere further out) and the v-axis at v = **±4.4 cm s⁻¹** (and nowhere further out). The v–x graph for s.h.m. is an ellipse, $v = \pm\omega\sqrt{(x_0^2 - x^2)}$, not straight lines between the intercepts.

(a)(ii) [2 marks]

$v_0 = \omega x_0$ and $\omega = 2\pi f$:
$f = v_0/(2\pi x_0) = 4.4/(2\pi \times 2.5) = 4.4/15.7$
= **0.28 Hz**

(b)(i) [2 marks]

Resonance is the oscillation of an object at **maximum amplitude**, which occurs when the **driving frequency equals the natural frequency** of the oscillating system.

(b)(ii) [2 marks]

Draw a curve rising continuously from f = 0 (amplitude never zero when f > 0) to a single peak at f = **0.28 Hz**, then falling continuously to f = 0.40 Hz.